



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY
LONERE, MAHARASHTRA, INDIA.
Department of Computer Science and Engineering

Neuromorphic Computing:

Hardware inspired by human brain

Presented by: Ammara Sirkhot

PRN: 24030332905059

Branch: Integrated Computer Science and Engineering

Class: Second Year/Third Semester

Under guidance: Prof. Uzma Munde

Content

- 1 What is Neuromorphic Computing?
- 2 How Does It Work?
- 3 The Magic Behind It
- 4 Applications
- 5 Neuromorphic Chips
- 6 Why Is It So Special?
- 7 Challenges

What is Neuromorphic Computing?

- Neuromorphic computing is a special type of computing that tries to work like the **human brain**.
- It processes information quickly, efficiently, and in parallel, like how our brain works.
- Neuromorphic systems are inspired by **neurons and synapses** in biological brains.

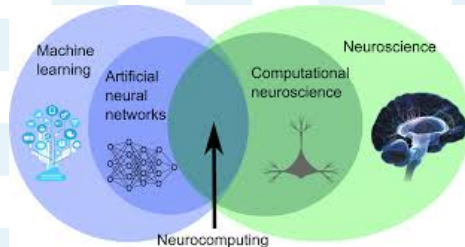
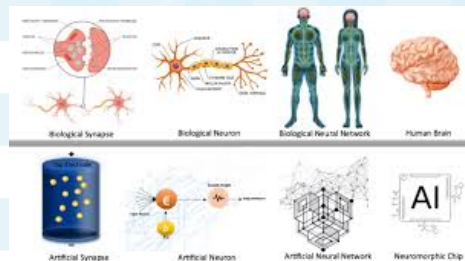


Figure: Neuro computing

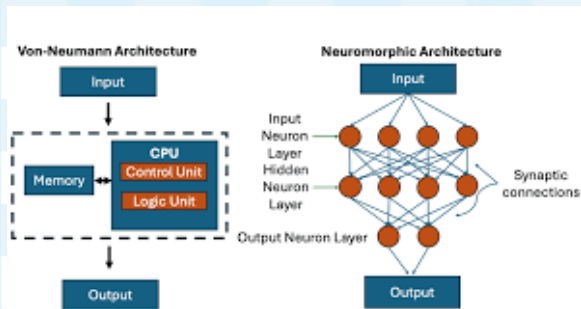
How Does It Work?

- Our brain is made up of billions of neurons connected by synapses.
- Neurons send tiny electrical signals to communicate with each other.
- **Neuromorphic chips** use artificial neurons that send electronic spikes, just like neurons in our brain.
- These spikes help the system learn and process information efficiently.



The Magic Behind Neuromorphic Computing

- Traditional computers do tasks one by one, which can be slow and power-hungry.
- Neuromorphic systems can handle many tasks at the same time (**parallel processing**).
- They react only when something happens, saving energy — just like our brain does.
- This makes them faster and more efficient than traditional systems for certain tasks.



Where Can We Use It?

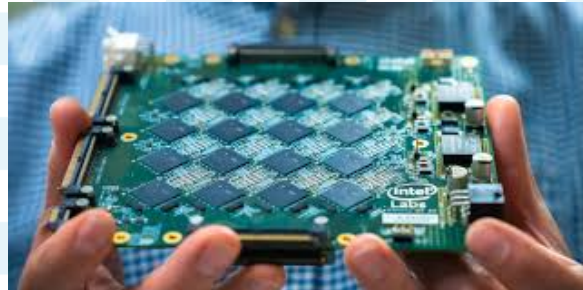
Neuromorphic computing is already making a difference in many fields:

- **Robotics:** For smarter, adaptive robot behaviour.
- **Healthcare:** In medical devices and brain–computer interfaces.
- **Self-driving cars:** For real-time decision making.
- **Drones:** For autonomous navigation.
- **Artificial Intelligence & Smart Devices:** For faster, energy-efficient AI.

Neuromorphic Chips

Some companies are already building amazing neuromorphic chips:

- ① **IBM TrueNorth:** Has 1 million neurons and can process huge amounts of data.
- ② **SpiNNaker:** Designed for super-fast learning and decision-making.
- ③ **BrainScaleS:** Focused on biologically realistic neural modeling.



Why Is It So Special?

- **Ultra Low Power:** Like the human brain, it requires very little energy compared to standard CPUs/GPUs.
- **Real-Time Processing:** It can process data instantly (in real-time) without sending it to the cloud (Edge Computing).
- **Fault Tolerance:** If one part fails, the system can adapt, similar to biological brains.
- **Compact:** High processing power in a small physical size.

Current Hurdles

- **Hardware Complexity:** Creating artificial synapses that perfectly mimic biological ones is difficult.
- **Software Compatibility:** We need new programming languages; standard code (like C++ or Python) doesn't work directly on these chips.
- **Cost:** Developing and manufacturing these specialized chips is currently very expensive.

Neuromorphic Computing vs. Artificial Intelligence

Not the same thing!

Artificial Intelligence (AI)

Software that learns and makes decisions using algorithms.

Runs on conventional processors such as CPUs, GPUs, and cloud systems.

Neuromorphic Computing

Hardware designed physically to think and structure like the brain.

Operates on specialized neuromorphic chips with artificial neurons and synapses that communicate through spikes.

When combined, they can create super-smart, adaptive machines that learn in real time!

Future of Neuromorphic Computing

- **Brain-Computer Interfaces (BCI):** Direct communication between the brain and external devices (e.g., controlling prosthetics).
- **Smarter Phones:** Mobile phones that can learn your habits offline without draining battery.
- **Cognitive Robotics:** Robots that can "feel" and interact with their environment safely.
- **Understanding the Brain:** Building these chips helps neuroscientists understand how the actual human brain functions.

Conclusion

- Neuromorphic computing combines neuroscience and technology.
- It's energy-efficient, intelligent, and adaptive.
- It represents the future of smart machines and brain-inspired computing.

“A step closer to machines that think like humans.”



References

-  M. S. Hasan, “An Overview of Neuromorphic Computing for Artificial Intelligence Enabled Hardware-Based Hopfield Neural Network,” 2023.
-  S. Furber *et al.*, “2022 Roadmap on Neuromorphic Computing and Engineering,” Neuromorphic Engineering, 2022.
-  J. Smith and R. Brown, “Brain-inspired computing systems: a systematic literature review,” Journal of Computer Systems, 2021.
-  P. Gupta, “Neuromorphic Computing and Applications,” Review Research Paper, 2020.